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**BBM475 Management Information Systems**  
**2025-26 Fall, Term Project on Business Process Modeling**  
**Due Dates: (T1) Nov 13, 2025, (T2: A & B) Dec 25, 2025**

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The purpose of this project is to introduce you with a real company (in terms of structure, behavior, information systems-IS, information technologies-IT, etc.) in relation to the topics covered by chapters 1-5 of your textbook. For the project, you are expected to find and contact a company that provides services in any business (cargo, polyclinic, restaurant, café and etc.), and analyze and define the concepts that you see there.

The project will consist of two stages and two deliveries at the end of these stages. As shown below, the first delivery (T1) will be evaluated over 100, the second delivery (T2) will be evaluated over 200, and your final grade will be calculated by dividing the sum of the two by 3:

- (T1) Definition and analysis of the current situation (**Nov 13, 100p**)
- (T2) Description of the targeted situation as a result of IS/IT infrastructure improvement proposals (**Dec 25, 200p**)

You can carry out the project in groups of 2-3 people. The roles and responsibilities of each student in the group should be clearly identified and defined in detail in the submissions.

>> Here is what you need to do to create **the first delivery (T1)**:

- 1)** Determine a workplace you will use in your project. Talk to company management and get its approval for this project.
- 2)** Take photos of group members at each business meeting and include them in your reports (along with date information) to document that each meeting takes place at the business location.
  - a. Each group member is not obliged to participate in each meeting. However, each member must attend at least one interview and be present in at least one photograph.
- 3)** In your first delivery report:
  - a. Describe the business, industry, features, basic services, product information and its brief history.
  - b. Define and model the process of **Receiving product/service orders** of the company.
  - c. Provide the technical details of the automation (device, software, communication, etc.) that support the ordering process of the company.
  - d. Explain how the business uses this automation in the ordering process (which steps the automation supports, and if applicable, subsidiary processes in relation to automation).
  - e. Explain how and why they moved from the old order receiving system to this new system with automation support. What were the reasons for this transition?
  - f. What did they make missing or wrong in the transition?
  - g. What would they do differently if they were to design the order system again?
  - h. What were the advantages of automation?
  - i. Use the following diagrams for the automation in question as you need, and add the graphs to your report with the necessary explanations:
    - i. Flowchart
    - ii. Business Process Model (with BPMN)
    - iii. Entity-Relationship (E-R) Diagram (to create a data model)
    - iv. Use Case Diagram (with UML)
    - v. Sequence Diagram (with UML)

>> Here is what you need to do to create **the second delivery (T2)**:

The second delivery will consist of two main parts: **A (TPS)** and **B (MIS)**.

*You can also make corrections in the first delivery (T1), add missing diagrams or parts, and retransmit with the second delivery, if you need at this stage. The first delivery you update will not be re-graded, but will provide better quality information as the basis for the evaluation of the second delivery.*

**(A) Transaction Processing System (TPS) - 100 points:**

- 1) Suggest a Transaction Processing System (TPS) for a process where the business you are reviewing will gain a Competitive Advantage over competitors.
- 2) Important issues to consider are given below:
  - a. Only **one new process** will be studied.
  - b. The proposed system will be TPS – will not be (Management Information System (MIS) or Decision Support System (DSS)!
  - c. It will not be one of the processes in the current operation!
  - d. The owner of the company (or the most authorized person in the workplace) must confirm that it can provide a competitive advantage (you may include what s/he thinks in the form of “interview notes” in your report.
- 3) Add the following to the section (A) of your report:
  - a. Explanation of the new process supported by TPS which will provide competitive advantage. (What it is, how it will enable competition, what data will be collected, what problems will be solved, what kind of competitive advantage it will bring, what the IS/IT infrastructure will be, what kind of data will be used, etc.)
  - b. For your newly proposed process supported by TPS, use the following diagrams as you need and add the graphs to your report with the necessary explanations.
    - i. Flowchart
    - ii. Entity-Relationship (E-R) diagram (to create a data model)
    - iii. Business Process Model (with BPMN)
    - iv. Use Case Diagram (with UML)
    - v. Sequence Diagram (with UML)
  - c. Demonstrate how the newly proposed process supported by TPS integrates with existing automation by referring to (or adding over) your prior diagrams.

**(B) Management Information System (MIS) – 100 points:**

- 1) Suggest a Management Information System (MIS) where the workplace you are studying will achieve a Competitive Advantage over competitors.
- 2) The MIS you propose should collect data from and influence both the Service and Production processes of the workplace.
- 3) Add the following to the section (B) of your report:
  - a. Explanation of the new process supported by MIS which will provide competitive advantage. (What it is, how it will enable competition, what data will be collected, what problems will be solved, what kind of competitive advantage it will bring, what the IS/IT infrastructure will be, what kind of data will be used, etc.)
  - b. Regarding your newly proposed process supported by MIS, use the following diagrams and add the graphs to your report with the necessary explanations.
    - i. Business Process Model (with BPMN)
    - ii. Entity-Relationship (E-R) diagram (to create a data model)
    - iii. Use Case Diagram (with UML)
  - c. Show how your newly proposed process supported by MIS will integrate with the process and the TPS automation you suggested in section (A) by referring to (or adding over) your prior diagrams.
- 4) Also add the following discussions to section (B) of your report:
  - a. Do the TPS and/or MIS that you have developed in T2 have dimensions to create ethical problems? If yes, what are they?
  - b. Is it reasonable to propose a Decision Support System (DSS) over the TPS and/or MIS that you have developed in T2? If yes, how and in what aspects could it be useful? Please explain in detail.

**References:**

- [1] Bizagi Modeler, <https://www.bizagi.com/en/platform/modeler>, for process modeling with BPMN.
- [2] Process Modeling Lecture Video by Marlon Dumas:  
<https://ut.cloud.panopto.eu/Panopto/Pages/Viewer.aspx?id=74d2e23e-34d6-4e15-9c6a-ab6c008bf25a>
- [3] Visual Paradigm, <https://www.visual-paradigm.com/download/community.jsp>, for software modeling with UML.